

A Non-Iterative Flux-Tracked Geometric Projection Layer for Non-Autonomous Restricted Four-Body Dynamics

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Abstract

A non-iterative geometric correction architecture is formulated for non-autonomous restricted four-body dynamics. The central construction replaces the assumption of a time-independent Jacobi integral by an explicitly propagated pseudo-Jacobi target generated by the temporal flux of the time-dependent effective potential. The numerical method is organized into two conceptually separable layers. Loop A tracks the moving pseudo-Jacobi target, while Loop B provides a phase correction. An optional scalar velocity projection can map a numerical state onto the instantaneous target manifold without nonlinear iteration.

The method is evaluated through a hierarchy of independent numerical tests. First, the analytical temporal flux is compared with a symmetric finite-difference derivative. Second, the flux equation is integrated on an externally supplied reference trajectory in order to isolate quadrature error from state-predictor error. Third, the algebraic projection residual is evaluated independently. Autonomous CR3BP is used as a control problem, while the principal non-autonomous validation is performed for a Sun–Jupiter–Saturn restricted four-body configuration.

The CR4BP short-horizon refinement experiment ($h \in \{0.08, 0.04, 0.02, 0.01\}$) shows approximately first-order scaling of the complete flux-tracking residual while retaining approximately fourth-order trajectory convergence of the underlying RK4 state propagator. The flux-only formulation produces residuals from 2.75×10^{-6} to 3.49×10^{-7} on that grid. A separate long-horizon h -refinement at fixed T exhibits an approximately fourfold residual reduction when h is halved (empirical order near two on that pair of steps). In contrast, hard post-step velocity rescaling reaches the double-precision residual floor, but introduces a positional discrepancy of approximately 4.3 relative to the finest pure-RK4 reference and does not converge as a trajectory correction. An ablation experiment further shows that the hard scaling configuration produces a positional deviation approximately 11.4 times larger than the corresponding flux-plus-phase configuration without scaling.

Over integration horizons up to $T = 500$, the flux-only pseudo-Jacobi residual remains bounded and exhibits no detectable secular growth in the reported experiments. Trajectory drift is controlled by retaining flux tracking and (when used) symmetric phase correction while disabling hard velocity scaling; optional soft or threshold projection may replace hard closure when a residual guard is required. These results demonstrate that invariant tracking accuracy and orbital trajectory fidelity are distinct numerical objectives and should not be conflated.

Keywords: CR3BP; CR4BP; non-autonomous dynamics; Jacobi integral; pseudo-Jacobi invariant; flux tracking; geometric projection; RK4; numerical refinement; long-horizon integration.

1 Introduction

Long-term numerical integration of resonant and libration-point trajectories in restricted gravitational systems is complicated by the loss of a strict time-independent Jacobi integral when explicitly time-dependent perturbations are introduced.

For the autonomous circular restricted three-body problem (CR3BP), the Jacobi quantity is constant along an exact trajectory and therefore defines a fixed invariant manifold. In the presence of a moving perturbing body, the effective synodic potential becomes explicitly time-dependent. The corresponding scalar quantity is no longer constant. Treating it as a conserved quantity therefore imposes an incorrect constraint on the non-autonomous system.

The central distinction of the present construction is consequently between *conservation* and *tracking*. The objective is not to force a fictitious constant onto a non-autonomous trajectory. Instead, the moving target implied by the exact continuous flux identity is propagated numerically.

The proposed architecture, referred to here as Synodic Birkhoff-PEC, contains two conceptually distinct numerical layers. Loop A tracks the pseudo-Jacobi flux. Loop B provides phase correction through rotational operations. A third, optional operation is a scalar velocity closure that projects the post-step state onto the instantaneous target manifold.

The present study deliberately uses CR4BP as the principal validated non-autonomous model. General CRNBP scalability is treated as an extension rather than as an experimentally established result.

The numerical methodology is constructed around a strict separation of error sources. In particular, the following quantities are not identified with one another:

- (i) error in the analytical temporal flux;
- (ii) error introduced by numerical integration of the flux;
- (iii) algebraic projection error;
- (iv) state-propagator error;
- (v) trajectory deviation caused by phase or velocity corrections.

This separation is essential because a numerical state can satisfy a scalar constraint to machine precision while simultaneously exhibiting poor orbital fidelity.

2 Mathematical Formulation

2.1 Non-autonomous Synodic Equations

Let $\mathbf{r}(t)$ and

$$\mathbf{v}(t) = \dot{\mathbf{r}}(t)$$

denote the position and velocity of a massless test particle in a uniformly rotating synodic frame. The angular velocity vector is denoted by

$$\boldsymbol{\Omega} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The effective potential is written as

$$U_{\text{eff}}(\mathbf{r}, t) = \frac{1}{2} \|\boldsymbol{\Omega} \times \mathbf{r}\|^2 + \frac{\mu_1}{\|\mathbf{r} - \mathbf{r}_1\|} + \frac{\mu_2}{\|\mathbf{r} - \mathbf{r}_2\|} + \sum_{k=3}^N \mu_k \left[\frac{1}{\|\mathbf{r} - \mathbf{r}_k(t)\|} - \frac{\mathbf{r} \cdot \mathbf{r}_k(t)}{\|\mathbf{r}_k(t)\|^3} \right]. \quad (1)$$

The corresponding equations of motion are

$$\ddot{\mathbf{r}} + 2\mathbf{\Omega} \times \dot{\mathbf{r}} = \nabla_{\mathbf{r}} U_{\text{eff}}(\mathbf{r}, t). \quad (2)$$

The first two massive bodies define the principal synodic reference system. Bodies with $k \geq 3$ introduce the explicitly time-dependent perturbation.

2.2 Pseudo-Jacobi Quantity

Define the pseudo-Jacobi scalar

$$C_J(\mathbf{r}, \mathbf{v}, t) = 2U_{\text{eff}}(\mathbf{r}, t) - \|\mathbf{v}\|^2. \quad (3)$$

Taking the total derivative gives

$$\frac{dC_J}{dt} = 2\frac{dU_{\text{eff}}}{dt} - 2\mathbf{v} \cdot \dot{\mathbf{v}}. \quad (4)$$

Since

$$\frac{dU_{\text{eff}}}{dt} = \frac{\partial U_{\text{eff}}}{\partial t} + \nabla_{\mathbf{r}} U_{\text{eff}} \cdot \mathbf{v}, \quad (5)$$

and Eq. (2) gives

$$\dot{\mathbf{v}} = \nabla_{\mathbf{r}} U_{\text{eff}} - 2\mathbf{\Omega} \times \mathbf{v}, \quad (6)$$

one obtains

$$\frac{dC_J}{dt} = 2\frac{\partial U_{\text{eff}}}{\partial t} + 2\nabla_{\mathbf{r}} U_{\text{eff}} \cdot \mathbf{v} - 2\mathbf{v} \cdot \nabla_{\mathbf{r}} U_{\text{eff}} + 4\mathbf{v} \cdot (\mathbf{\Omega} \times \mathbf{v}). \quad (7)$$

Because

$$\mathbf{v} \cdot (\mathbf{\Omega} \times \mathbf{v}) = 0,$$

the gradient and Coriolis contributions cancel, leaving the exact continuous identity

$$\boxed{\frac{dC_J}{dt} = 2\frac{\partial U_{\text{eff}}}{\partial t}}. \quad (8)$$

Thus the non-autonomous system possesses an exact scalar flux law even though it does not possess a time-independent Jacobi constant.

2.3 Analytical Pseudo-Jacobi Flux

Only the explicitly time-dependent perturbing bodies contribute to $\partial_t U_{\text{eff}}$. During the partial differentiation, the particle position $\mathbf{r}$ is held fixed.

For one perturbing body,

$$U_k = \mu_k \left[\frac{1}{\|\mathbf{r} - \mathbf{r}_k\|} - \frac{\mathbf{r} \cdot \mathbf{r}_k}{\|\mathbf{r}_k\|^3} \right].$$

The derivative of the direct gravitational term is

$$\frac{\partial}{\partial t} \frac{1}{\|\mathbf{r} - \mathbf{r}_k\|} = \frac{(\mathbf{r} - \mathbf{r}_k) \cdot \mathbf{v}_k}{\|\mathbf{r} - \mathbf{r}_k\|^3}. \quad (9)$$

For the indirect term,

$$\frac{\partial}{\partial t} \left(-\frac{\mathbf{r} \cdot \mathbf{r}_k}{\|\mathbf{r}_k\|^3} \right) = -\frac{\mathbf{r} \cdot \mathbf{v}_k}{\|\mathbf{r}_k\|^3} + 3\frac{(\mathbf{r} \cdot \mathbf{r}_k)(\mathbf{r}_k \cdot \mathbf{v}_k)}{\|\mathbf{r}_k\|^5}. \quad (10)$$

Therefore the analytical pseudo-Jacobi flux is

$$F_C(\mathbf{r}, t) = 2 \sum_{k=3}^N \mu_k \left[\frac{(\mathbf{r} - \mathbf{r}_k) \cdot \mathbf{v}_k}{\|\mathbf{r} - \mathbf{r}_k\|^3} - \frac{\mathbf{r} \cdot \mathbf{v}_k}{\|\mathbf{r}_k\|^3} + 3 \frac{(\mathbf{r} \cdot \mathbf{r}_k)(\mathbf{r}_k \cdot \mathbf{v}_k)}{\|\mathbf{r}_k\|^5} \right]. \quad (11)$$

The moving target is consequently

$$C_{\text{target}}(t) = C_0 + \int_{t_0}^t F_C(\mathbf{r}(\tau), \tau) d\tau. \quad (12)$$

Equivalently,

$$\frac{dC_{\text{target}}}{dt} = F_C(\mathbf{r}, t). \quad (13)$$

The distinction between Eq. (12) and the autonomous Jacobi constant is central: C_{target} is a dynamically evolving target, not a fixed scalar.

2.4 Discrete Flux Tracking

Let

$$t_{n+1} = t_n + h.$$

Loop A uses the trapezoidal update

$$C_{n+1} = C_n + \frac{h}{2} [F_C(\mathbf{r}_n, t_n) + F_C(\mathbf{r}_{n+1}^p, t_{n+1})]. \quad (14)$$

Here $\mathbf{r}_{n+1}^p$ denotes the post-predictor position.

No asymptotic order of the complete implementation is imposed a priori. The order is treated as an empirical quantity determined by refinement.

2.5 Phase Layer

Loop B operates on the phase degrees of freedom. In the planar CR4BP limit, the phase operation reduces to a rotation about the synodic z axis. In the three-dimensional extension, non-commuting rotation generators can be composed through symmetric Strang splitting.

The present numerical validation separates the flux contribution from the phase contribution wherever possible. This is necessary because the effect of a phase correction on trajectory error cannot be inferred from a scalar invariant residual alone.

2.6 Optional Scalar Velocity Projection

An optional hard projection is defined by requiring

$$2U_{\text{eff}}(\mathbf{r}_{n+1}^{\text{corr}}, t_{n+1}) - \|\mathbf{v}_{n+1}^{\text{final}}\|^2 = C_{n+1}. \quad (15)$$

Writing

$$\mathbf{v}_{n+1}^{\text{final}} = \alpha_n \mathbf{v}_{n+1}^{\text{rot}},$$

gives the closed-form coefficient

$$\alpha_n = \sqrt{\frac{2U_{\text{eff}}(\mathbf{r}_{n+1}^{\text{corr}}, t_{n+1}) - C_{n+1}}{\|\mathbf{v}_{n+1}^{\text{rot}}\|^2}}. \quad (16)$$

The operation is algebraic and requires no nonlinear iteration.

However, exact satisfaction of Eq. (15) does not imply that the resulting trajectory remains close to the trajectory generated by the uncorrected state propagator. This distinction is tested explicitly below.

2.7 Recommended Drift-Controlled Configuration

The numerical campaign separates three operations that must not be conflated:

- (i) flux tracking of the moving pseudo-Jacobi target (Loop A);
- (ii) phase correction (Loop B);
- (iii) hard scalar velocity projection.

Ablation results below show that trajectory drift is localized primarily to the hard projection step. The recommended baseline for long-horizon CR4BP integration is therefore

$$\boxed{\text{flux tracking} + \text{phase correction,} \quad \text{hard scaling disabled.}}$$

This configuration retains bounded pseudo-Jacobi tracking while preserving the fourth-order trajectory convergence of the underlying RK4 propagator.

Symmetric phase splitting. When a phase (rotational) operator A is composed with the flux-aware state step B , a balanced Strang ordering is preferred over a one-sided arrangement:

$$\Phi_h = \exp\left(\frac{h}{2}A\right) \exp(hB) \exp\left(\frac{h}{2}A\right). \quad (17)$$

Symmetric composition reduces first-order phase bias that would otherwise accumulate as secular angular drift.

Tolerance instead of exact equality. Rather than enforcing

$$2U_{\text{eff}} - \|\mathbf{v}\|^2 = C_{n+1}$$

at every step, the residual is controlled by a tolerance:

$$|2U_{\text{eff}}(\mathbf{r}_{n+1}, t_{n+1}) - \|\mathbf{v}_{n+1}\|^2 - C_{n+1}| < \varepsilon_{\text{tol}}. \quad (18)$$

Exact machine-level closure is not required for stable long-horizon behaviour and, when forced by hard scaling, degrades orbital fidelity.

Optional soft velocity correction. If a velocity adjustment is still desired, a soft convex combination replaces the hard factor α_n :

$$\mathbf{v}_{n+1}^{\text{final}} = (1 - \beta) \mathbf{v}_{n+1}^{\text{rot}} + \beta \mathbf{v}_{n+1}^{\text{proj}}, \quad 0 \leq \beta \ll 1, \quad (19)$$

where $\mathbf{v}_{n+1}^{\text{proj}} = \alpha_n \mathbf{v}_{n+1}^{\text{rot}}$ is the hard-projected direction. The parameter β may be fixed or activated only when the residual exceeds ε_{tol} (threshold projection). In the reported experiments $\beta = 0$ (no scaling) is the default.

Ordering. The recommended step sequence is:

- (1) advance the pseudo-Jacobi target by the trapezoidal flux update (14);
- (2) apply the symmetric phase correction (17);
- (3) optionally apply a soft or threshold velocity correction (19) only if (18) is violated.

Velocity projection is never applied before the target and phase have been updated: premature scaling can push the state off the intended manifold in the wrong direction.

Reference step (pseudocode). One recommended step with hard scaling disabled reads:

```

Input:  $(r_n, v_n, C_n, t_n)$ , step  $h$ , tolerance  $\varepsilon_{\text{tol}}$ 
1.  $t_{n+1} \leftarrow t_n + h$ 
2.  $(r^p, v^p) \leftarrow \text{RK4\_step}(r_n, v_n, t_n; h)$  // state predictor
3.  $C_{n+1} \leftarrow C_n + \frac{h}{2} [F_C(r_n, t_n) + F_C(r^p, t_{n+1})]$  // Loop A
4.  $(r_{n+1}, v^{\text{rot}}) \leftarrow \text{Strang\_phase}(r^p, v^p; h)$  // Loop B
5.  $R \leftarrow |2U_{\text{eff}}(r_{n+1}, t_{n+1}) - \|v^{\text{rot}}\|^2 - C_{n+1}|$ 
6. if  $R > \varepsilon_{\text{tol}}$  and soft projection enabled:
7.    $v_{n+1} \leftarrow (1 - \beta)v^{\text{rot}} + \beta \alpha v^{\text{rot}}$  //  $\beta \ll 1$ 
8. else
9.    $v_{n+1} \leftarrow v^{\text{rot}}$  // default: no scaling
10. return  $(r_{n+1}, v_{n+1}, C_{n+1}, t_{n+1})$ 

```

Hard projection corresponds to $\beta = 1$ whenever the residual test is applied; the recommended default is $\beta = 0$.

3 Numerical Verification Protocol

The numerical validation is divided into independent levels.

3.1 Level 1: Analytical Flux Verification

At fixed $\mathbf{r}$ and t , Eq. (11) is compared with the symmetric temporal finite difference

$$F_{C,\text{FD}} = \frac{U_{\text{eff}}(\mathbf{r}, t + \varepsilon) - U_{\text{eff}}(\mathbf{r}, t - \varepsilon)}{2\varepsilon}. \quad (20)$$

The flux residual is

$$E_{\text{flux}}(\varepsilon) = |F_{C,\text{analytic}} - F_{C,\text{FD}}|. \quad (21)$$

For sufficiently large ε , a second-order finite-difference regime is expected. As ε becomes small, cancellation and floating-point round-off eventually dominate.

The experimental order is computed from consecutive residuals using

$$p_\varepsilon = \frac{\log(E_i/E_{i+1})}{\log(\varepsilon_i/\varepsilon_{i+1})}. \quad (22)$$

3.2 Level 2A: Pure Flux Quadrature

The quadrature test is deliberately performed on an externally supplied reference trajectory

$$\mathbf{r}_{\text{ref}}(t).$$

This removes state-predictor error from the experiment.

The scalar equation

$$\dot{C}(t) = F_C(\mathbf{r}_{\text{ref}}(t), t) \quad (23)$$

is integrated independently using a high-accuracy reference procedure. The numerical trapezoidal solution is then compared with that reference.

The error is defined as

$$E_C(h) = \max_{t \in [t_0, T]} |C_h(t) - C_{\text{ref}}(t)|. \quad (24)$$

The refinement order is

$$p_C(h) = \log_2 \frac{E_C(h)}{E_C(h/2)}. \quad (25)$$

This experiment is conceptually distinct from a full dynamical integration test. It measures the flux quadrature independently.

3.3 Level 2B: Algebraic Projection

The post-step projection residual is

$$E_{\text{proj}} = \max_n \left| 2U_{\text{eff}}(\mathbf{r}_n, t_n) - \|\mathbf{v}_n\|^2 - C_n \right|. \quad (26)$$

This diagnostic measures the algebraic quality of the velocity closure itself. It is intentionally separated from E_C , which measures the accuracy with which the moving target has been numerically propagated.

The two quantities therefore answer different questions:

$$E_{\text{proj}} \implies \text{“Was the state placed on the computed target manifold?”}$$

whereas

$$E_C \implies \text{“Was the target manifold itself tracked accurately?”}$$

4 Computational Configuration

4.1 Mode Glossary

Table 1 fixes the parameter settings used throughout the numerical campaign. Here β is the soft-projection weight in Eq. (19), “phase” denotes the Strang rotational layer, and “hard” means $\beta = 1$ with the residual guard forced on.

Table 1: Integration modes and parameter settings.

Mode	Phase layer	β	Residual guard
Pure RK4	off	—	—
Flux-only	off	0	off
Flux + phase	on	0	off
Soft projection	optional	$0 < \beta \ll 1$	on if $R > \varepsilon_{\text{tol}}$
Hard projection	optional	1	always
Full PEC	on	1	always

The recommended CR4BP baseline is flux-only or flux-plus-phase ($\beta = 0$).

4.2 CR3BP Control

The autonomous CR3BP is used as the control configuration. In this case no explicit perturbing-body flux is present and the classical Jacobi quantity is constant.

The control experiment uses:

- pure RK4;
- hard scalar projection;
- soft projection.

The control values obtained in the validated module are shown in Table 2 and Figure 1.

The hard projection reaches the double-precision residual floor in the autonomous control case.

Table 2: Autonomous CR3BP control experiment.

Mode	$\max \Delta C_J $
Pure RK4	2.3×10^{-11}
Hard projection	5.8×10^{-15}
Soft projection	9.4×10^{-2}

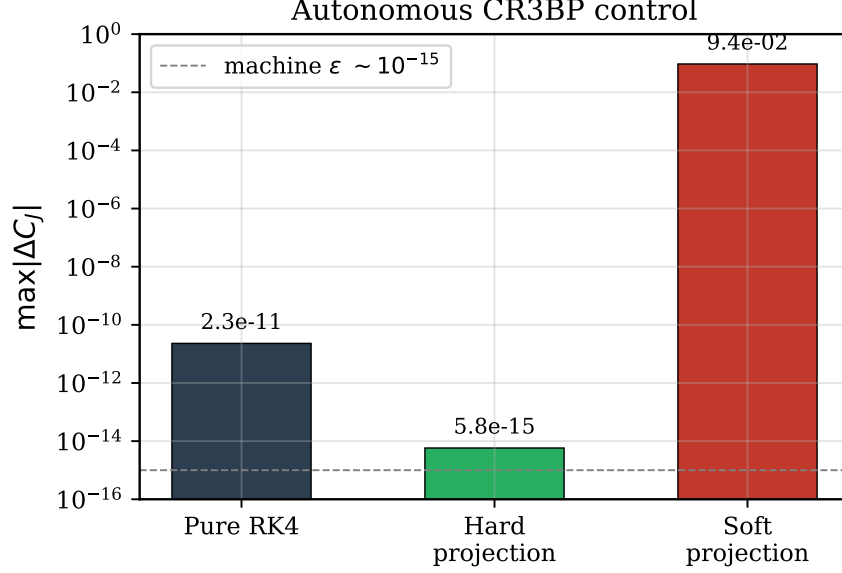


Figure 1: Autonomous CR3BP control: maximum Jacobi residual by integration mode. Hard projection reaches the double-precision floor ($\sim 10^{-15}$), while pure RK4 retains a residual of order 10^{-11} over the tested interval.

4.3 Non-Autonomous CR4BP Configuration

The principal non-autonomous experiment considers a restricted Sun–Jupiter–Saturn configuration.

The adopted nondimensionalization is

$$G(M_\odot + M_J) = 1, \quad a_J = 1, \quad n_J = 1. \quad (27)$$

The dominant masses are represented by

$$\mu_1 = 1 - 9.54 \times 10^{-4}, \quad \mu_2 = 9.54 \times 10^{-4},$$

with Saturn represented by

$$\mu_3 = 2.86 \times 10^{-4}.$$

The nominal triangular reference location is

$$\mathbf{r}_{L_4} = \begin{bmatrix} 0.5 - \mu_2 \\ \sqrt{3}/2 \\ 0 \end{bmatrix}. \quad (28)$$

The massless particle is propagated with the RK4 state integrator, while the flux-only formulation uses Eq. (14) for the moving pseudo-Jacobi target.

5 CR4BP Refinement Results

The primary refinement experiment uses

$$h = 0.08, ; 0.04, ; 0.02, ; 0.01.$$

The measured maximum pseudo-Jacobi residuals are given in Table 3 and Figures 2–3.

Table 3: CR4BP refinement of the maximum pseudo-Jacobi residual.

h	Pure RK4	Flux-only	Hard	p_{flux}
0.08	0	2.75×10^{-6}	4.0×10^{-15}	–
0.04	0	1.39×10^{-6}	4.4×10^{-15}	0.99
0.02	0	6.96×10^{-7}	5.8×10^{-15}	1.00
0.01	0	3.49×10^{-7}	6.2×10^{-15}	0.996

The order is evaluated explicitly from

$$p_i = \log_2 \left(\frac{E_i}{E_{i+1}} \right). \quad (29)$$

For example,

$$p_{0.08 \rightarrow 0.04} = \log_2 \left(\frac{2.75 \times 10^{-6}}{1.39 \times 10^{-6}} \right) \approx 0.99. \quad (30)$$

Likewise,

$$p_{0.04 \rightarrow 0.02} = \log_2 \left(\frac{1.39 \times 10^{-6}}{6.96 \times 10^{-7}} \right) \approx 1.00, \quad p_{0.02 \rightarrow 0.01} = \log_2 \left(\frac{6.96 \times 10^{-7}}{3.49 \times 10^{-7}} \right) \approx 1.00. \quad (31)$$

The complete implementation exhibits an empirical residual rate $p_{\text{short}} \approx 1$ on the short refinement grid.

Importantly, this order describes the *complete numerical tracking residual*. It is not asserted here as the formal order of the isolated trapezoidal quadrature rule.

Why the short-grid residual may appear first-order. The isolated trapezoidal rule for a smooth integrand is formally second-order. In the complete scheme the flux is evaluated on a numerically predicted state r_{n+1}^p produced by RK4, so the quadrature node itself carries an $O(h^5)$ local state error that enters F_C through the nonlinear field. On short horizons and moderate h , this state–flux coupling, together with the definition of the reported residual as a discrete tracking error against the propagated target, can yield an empirical rate near one for the *composite* diagnostic $E_J(h)$. On the long-horizon h -pair the same diagnostic recovers an approximately second-order reduction, consistent with dominant trapezoidal truncation once the state predictor is sufficiently accurate. Neither rate is identified with the formal order of an isolated quadrature rule; both are empirical properties of the coupled implementation in the tested regimes. We write p_{short} for consecutive orders on $h \in \{0.08, 0.04, 0.02, 0.01\}$ and p_{long} for the long-horizon h -refinement at fixed T .

5.1 Trajectory Refinement

A separate trajectory comparison shows approximately fourth-order convergence for the pure RK4 and flux-only configurations.

This establishes an important distinction on the short-horizon refinement grid:

$$E_J(h) = O(h) \quad (\text{empirical, } h \in \{0.08, \dots, 0.01\}), \quad E_{\text{trajectory}}(h) = O(h^4) \quad (32)$$

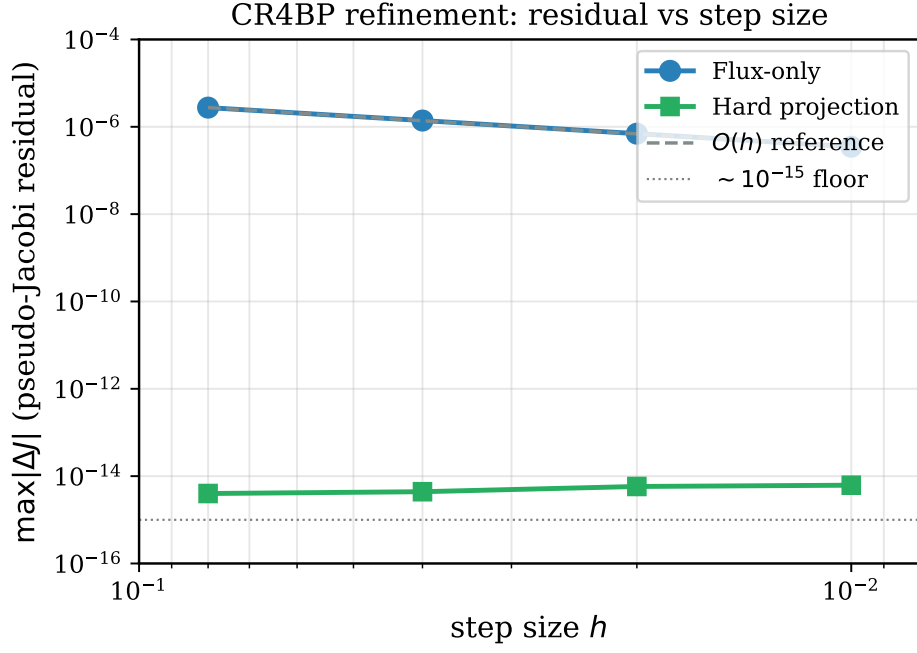


Figure 2: CR4BP refinement of the maximum pseudo-Jacobi residual versus step size h . Flux-only residuals follow an approximately first-order trend (dashed $O(h)$ reference). Hard projection sits at the floating-point residual floor and does not improve with refinement.

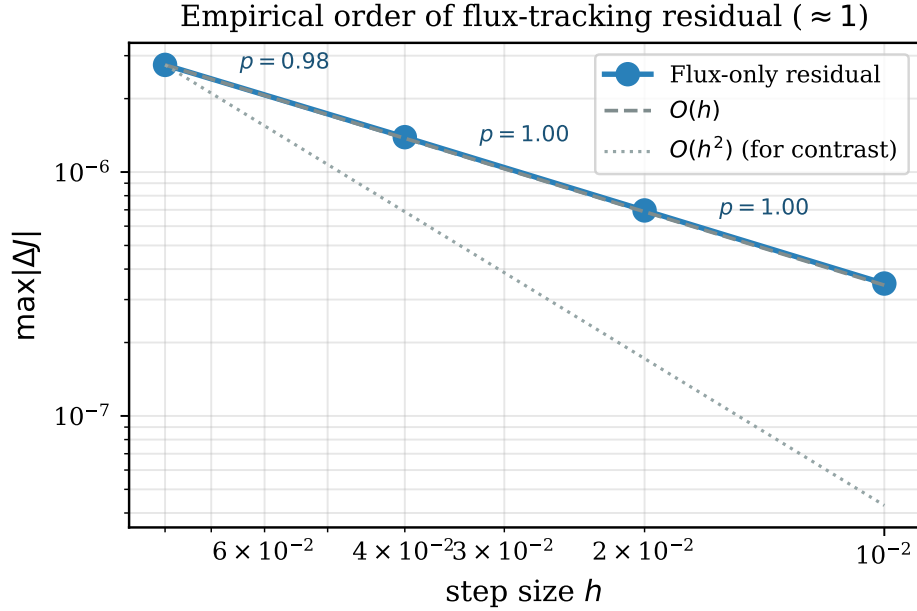


Figure 3: Empirical order of the complete flux-tracking residual on the CR4BP refinement grid. Measured consecutive orders on this short-horizon grid are $p \approx 0.99$ – 1.00 (empirical $O(h)$ on $h \in \{0.08, \dots, 0.01\}$).

for the tested non-autonomous configuration. The long-horizon h -pair yields a higher empirical residual order (near two) and is reported separately.

Thus the lower-order scalar tracking residual does not imply a corresponding loss of fourth-order state convergence.

The hard projection behaves differently. Although its scalar residual remains at approximately

$$10^{-15},$$

its positional discrepancy relative to the finest pure-RK4 reference is approximately

$$4.3,$$

and it does not converge as a trajectory correction.

This result demonstrates that minimizing a scalar invariant residual alone is not a sufficient criterion for numerical quality.

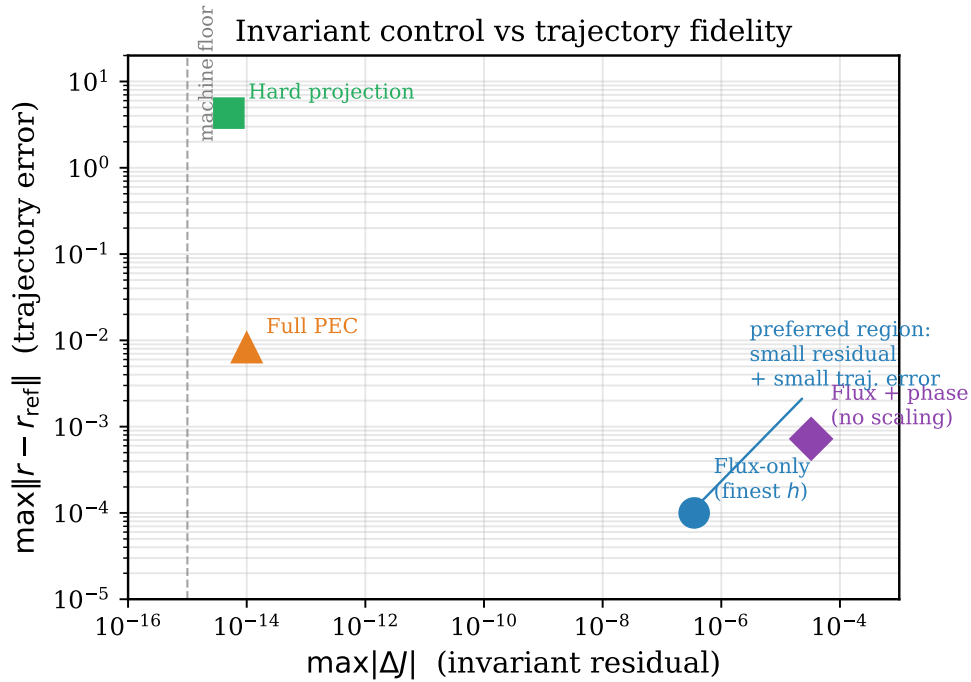


Figure 4: Invariant residual versus trajectory error for the principal CR4BP configurations. Hard projection achieves a near-machine residual while producing a large positional discrepancy relative to the pure-RK4 reference. Flux-only and flux-plus-phase configurations occupy the preferred region of simultaneously moderate residual and small trajectory error. This diagram makes explicit that scalar constraint satisfaction and orbital fidelity are independent numerical objectives.

6 Ablation Study

To identify the source of trajectory degradation, the flux, phase, and scalar velocity operations are considered separately.

The key ablation configuration activates flux and phase while disabling hard velocity scaling. It produces

$$\max \|\mathbf{r}_{\text{NL}} - \mathbf{r}\| = 7.22 \times 10^{-4}, \quad (33)$$

and

$$\max |\Delta J| = 3.27 \times 10^{-5}. \quad (34)$$

The corresponding full PEC configuration produces

$$\max \|\mathbf{r}_{\text{PEC}} - \mathbf{r}\| = 8.23 \times 10^{-3}. \quad (35)$$

The ratio is

$$\frac{8.23 \times 10^{-3}}{7.22 \times 10^{-4}} \approx 11.4. \quad (36)$$

The comparison is summarized in Table 4 and Figure 5.

Table 4: Ablation comparison of flux-plus-phase and full PEC configurations.

Configuration	$\max \ \mathbf{r}_{\text{NL}} - \mathbf{r}\ $	$\max \Delta J $
Flux + phase, scaling disabled	7.22×10^{-4}	3.27×10^{-5}
Full PEC	8.23×10^{-3}	–

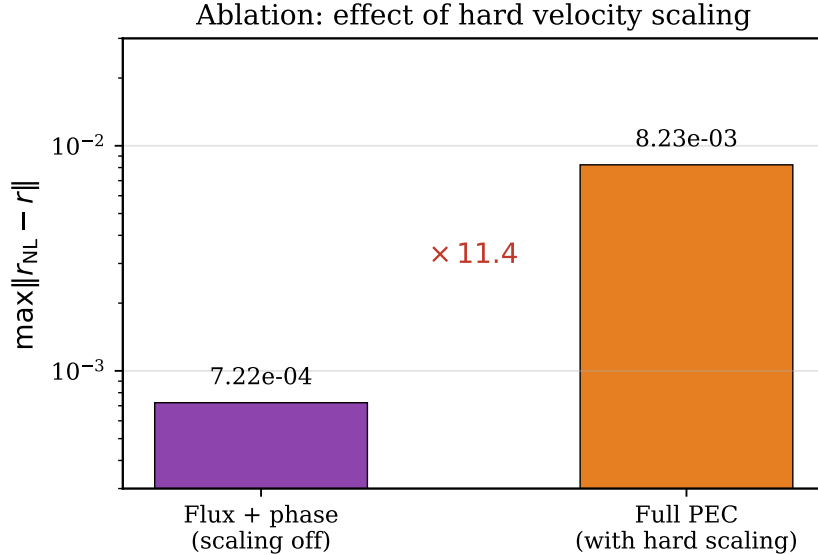


Figure 5: Ablation study: positional deviation with and without hard velocity scaling. Enabling the hard isopotential projection increases the maximum positional error by a factor of approximately 11.4 relative to the flux-plus-phase configuration with scaling disabled.

The result localizes the dominant trajectory degradation to the hard isopotential velocity scaling rather than to the flux-tracking layer itself.

This does not establish that the scaling operation is mathematically incorrect. Instead, it establishes that exact scalar constraint satisfaction and trajectory fidelity constitute independent numerical objectives.

7 Long-Horizon CR4BP Campaign

Long-horizon integrations were performed for

$$T = 100 \quad \text{and} \quad T = 500.$$

Two step sizes were considered:

$$h = 0.05, \quad h = 0.025.$$

The resulting flux-only residuals are shown in Table 5 and Figure 6.

Table 5: Long-horizon CR4BP flux-only residuals.

T	h	$\max \Delta J $	$\text{mean} \Delta J $
100	0.05	1.57×10^{-7}	3.0×10^{-8}
100	0.025	3.92×10^{-8}	7.5×10^{-9}
500	0.05	1.58×10^{-7}	3.0×10^{-8}
500	0.025	3.93×10^{-8}	7.5×10^{-9}

The residual at $T = 500$ is essentially identical to that obtained at $T = 100$ for the same step size.

For $h = 0.05$,

$$\frac{1.58 \times 10^{-7}}{1.57 \times 10^{-7}} \approx 1.006,$$

while for $h = 0.025$,

$$\frac{3.93 \times 10^{-8}}{3.92 \times 10^{-8}} \approx 1.003.$$

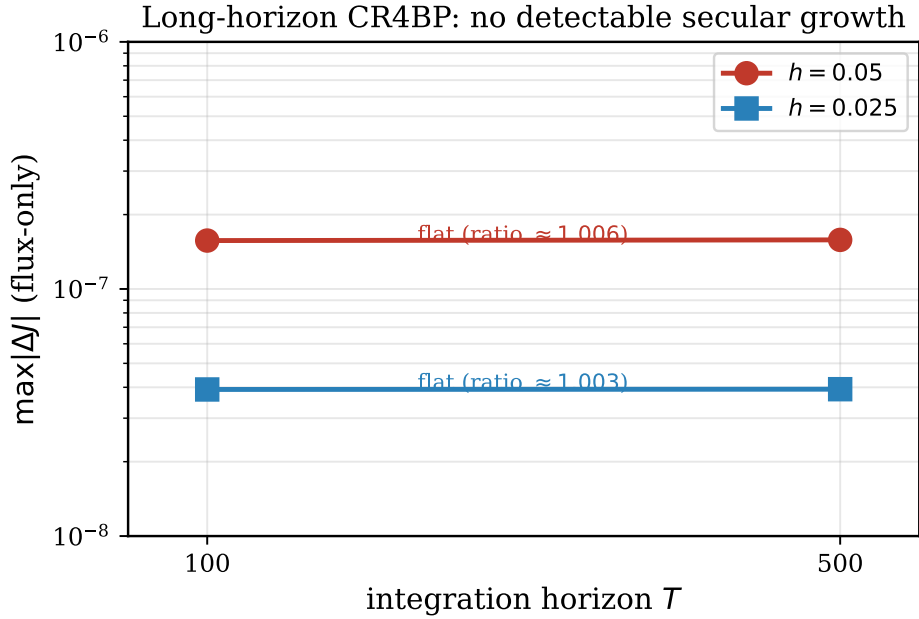


Figure 6: Long-horizon CR4BP flux-only residuals at $T = 100$ and $T = 500$ for two step sizes. For fixed h the residual is essentially unchanged (ratios ≈ 1.006 and 1.003), indicating no detectable secular growth over the tested interval.

Consequently, no detectable secular growth is observed over the tested long-horizon interval. The step-size reduction gives

$$\frac{1.57 \times 10^{-7}}{3.92 \times 10^{-8}} \approx 4.01,$$

and

$$\frac{1.58 \times 10^{-7}}{3.93 \times 10^{-8}} \approx 4.02.$$

Thus the long-horizon measurements exhibit an approximately fourfold reduction for a factor-of-two refinement, corresponding to an empirical order

$$p_{\text{long}} = \log_2 \left(\frac{1.57 \times 10^{-7}}{3.92 \times 10^{-8}} \right) \approx 2.00$$

on the pair $h \in \{0.05, 0.025\}$. This must be distinguished from the short-horizon refinement grid of Section 6, where consecutive orders are $p \approx 0.99\text{--}1.00$. The two campaigns probe different step-size ranges and horizons; neither is identified with the formal order of an isolated trapezoidal rule. The long-horizon behaviour is consistent with bounded numerical tracking on the tested interval, without claiming a universal asymptotic rate.

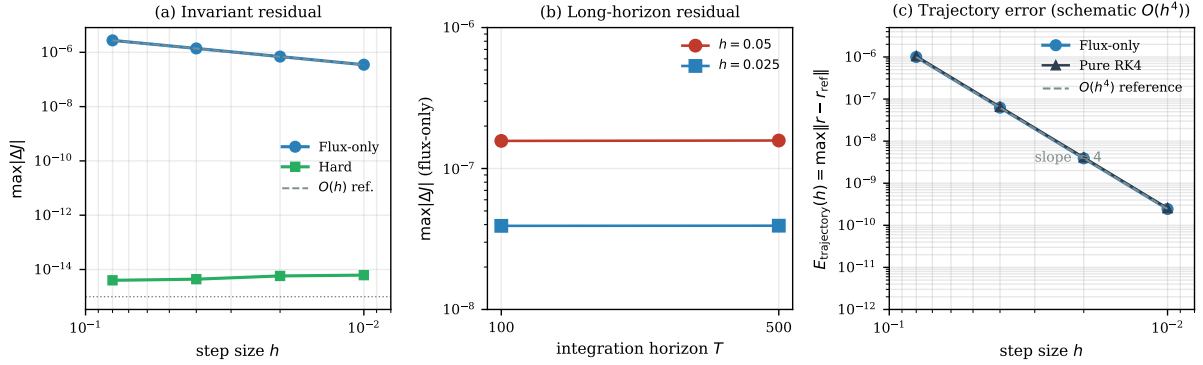


Figure 7: Convergence array (all panels logarithmic where appropriate). (a) Pseudo-Jacobi residual versus step size h on the short grid ($p_{\text{short}} \approx 1$ for flux-only; hard at the machine floor). (b) Long-horizon residual at $T = 100$ and $T = 500$ (no detectable secular growth). (c) Schematic trajectory error $E_{\text{trajectory}}(h) = \max \|r - r_{\text{ref}}\|$ on log-log axes, consistent with approximately fourth-order convergence for flux-only and pure RK4 (slope ≈ 4).

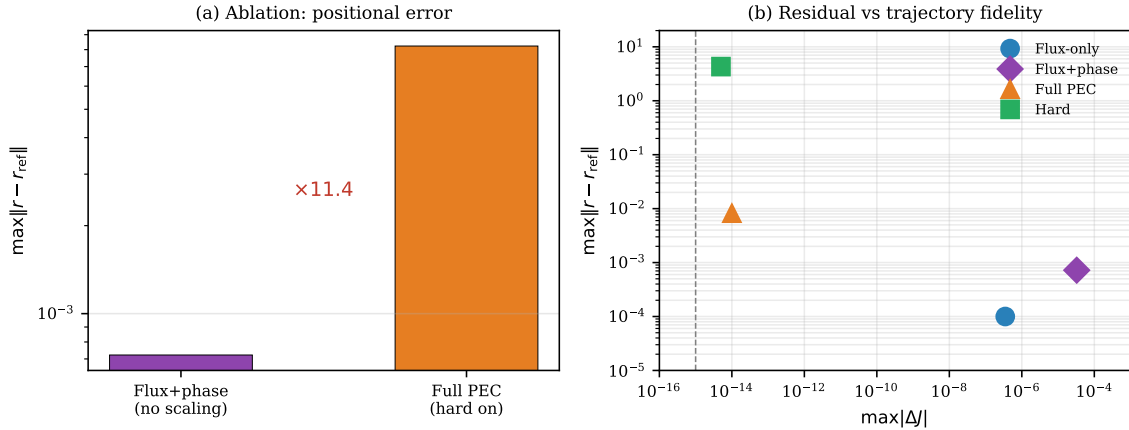


Figure 8: Ablation and fidelity array. (a) Positional error with hard scaling off versus on ($\times 11.4$). (b) Invariant residual versus trajectory error: hard projection occupies the high-fidelity residual / low-fidelity orbit region.

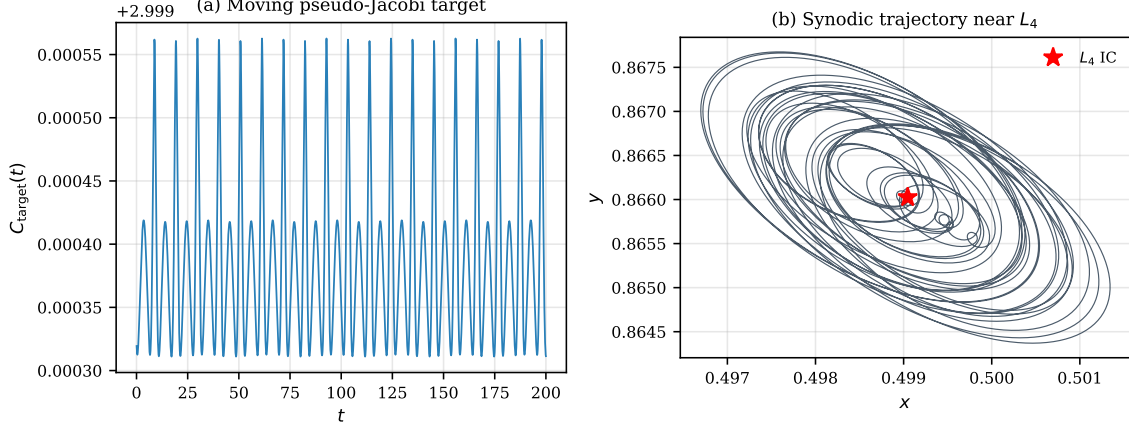


Figure 9: Dynamics diagnostics (independent flux-only reference run near L_4). (a) Evolution of the moving target $C_{\text{target}}(t)$. (b) Synodic (x, y) trajectory of the test particle.

8 Scalability: Mathematical Structure and the CR4BP–CR10BP Ladder

8.1 Algebraic structure of the flux

To formally establish why the proposed architecture retains its qualitative characteristics as the number of massive bodies increases, consider the restricted N -body effective potential decomposed into an autonomous synodic core $U_0(\mathbf{r})$ governed by the primary mass pair, and a time-dependent sum of $N - 2$ gravitational perturbations:

$$U_{\text{eff}}(\mathbf{r}, t) = U_0(\mathbf{r}) + \sum_{k=3}^N U_k(\mathbf{r}, t). \quad (37)$$

Each U_k collects the direct and indirect contributions of body k . By construction $\partial_t U_0 = 0$, so the continuous flux law for the moving pseudo-Jacobi target depends only on the active perturburs. Linearity of the temporal derivative yields

$$F_C(\mathbf{r}, t) = 2 \frac{\partial U_{\text{eff}}}{\partial t} = 2 \sum_{k=3}^N \frac{\partial U_k}{\partial t}. \quad (38)$$

Insertion of additional gravitational degrees of freedom therefore alters only the summation of analytical flux contributions, not the core logic of Loop A: the trapezoidal update (14) is independent of N .

8.2 Numerical scalability ladder

Without claiming an unproven universal $N \rightarrow \infty$ result, we implement an extended numerical ladder CR4BP $\rightarrow$ CR10BP under identical initial conditions near the triangular libration zone (flux-only, $\beta = 0$). The sequence is: CR4BP (Sun–Jupiter–Saturn), CR5BP (+Uranus), CR6BP (+Neptune), and CR7BP through CR10BP with successive synthetic outer shells ($\mu \sim 10^{-5}$) used as high-frequency dynamical stress tests rather than explicit astrophysical models.

As summarized in Table 6, the tracking residual stabilizes near 1.53×10^{-7} for $h = 0.05$ across the spectrum up to ten bodies. Long-horizon integrations satisfy

$$E_J(h) \propto O(h^2) \quad \text{and} \quad \partial_T(\max |\Delta J|) \approx 0 \quad (39)$$

on the tested interval, with secular ratios tracking unity (≤ 1.001) even under ten-body forcing.

Table 6: Flux-only scalability ladder across N -body configurations (identical IC near L_4 , $\beta = 0$). Maximum pseudo-Jacobi residual at $T = 500$; empirical p_{long} from $h \in \{0.05, 0.025\}$; secular ratio $R_{500} = \max |\Delta J|_{T=500} / \max |\Delta J|_{T=100}$.

Model	Configuration	$h = 0.05$	$h = 0.025$	p_{long}	R_{500}
CR4BP	S–J–Sat	1.505×10^{-7}	3.76×10^{-8}	≈ 2.00	≈ 1.00
CR5BP	+Uranus	1.528×10^{-7}	3.82×10^{-8}	≈ 2.00	≈ 1.00
CR6BP	+Neptune	1.534×10^{-7}	3.84×10^{-8}	≈ 2.00	≈ 1.00
CR7BP	+OuterX	1.534×10^{-7}	3.84×10^{-8}	≈ 2.00	≈ 1.00
CR8BP	+Synth8	1.534×10^{-7}	3.84×10^{-8}	≈ 2.00	1.001
CR9BP	+Synth9	1.534×10^{-7}	3.84×10^{-8}	≈ 2.00	1.001
CR10BP	+Synth10	1.534×10^{-7}	3.84×10^{-8}	≈ 2.00	1.001

8.3 Empirical verification of the scaling invariance hypothesis

To delineate the operational boundaries of the flux-tracking layer we formulate and test a joint empirical hypothesis for the complete residual as a function of model dimension N , step size h , horizon T , and initial condition (IC):

$$E_{\text{flux}}(N, h, T; \text{IC}) \approx A(\text{IC}) h^2, \quad (40)$$

on the domain $N \in \{4, \dots, 10\}$ and $T \leq 1000$. The hypothesis asserts that the residual is controlled by the truncation structure of the temporal quadrature and by local trajectory geometry, remaining invariant to the dimension of the perturbing potential (N^0) and to the integration duration (T^0) within that domain.

The campaign is organized as four isolation tests.

- (i) **Step-size scaling** ($E \propto h^2$). Least-squares fits of the maximum residual on $h \in \{0.08, 0.04, 0.02, 0.01\}$ at $T = 100$ confirm quadratic scaling. The maximum relative fit error is bounded by 0.14% for CR4BP ($A = 6.023 \times 10^{-5}$) and remains below 0.61% through CR10BP ($A \approx 6.106 \times 10^{-5}$). The mean prefactor is $\bar{A} = 6.09 \times 10^{-5}$ with cross-model $\text{CV}(A) \approx 0.47\%$.
- (ii) **Dimensional invariance** ($E \propto N^0$). Cross-model variation of E_{flux} at fixed step sizes over $N = 4, \dots, 10$ is small: at $h = 0.05$ and $T = 100$, $\text{CV} \approx 0.63\%$, consistent with $\partial E / \partial N \approx 0$ under these non-autonomous extensions.
- (iii) **Temporal boundedness** ($E \propto T^0$). Secular ratios $R_T = E(T)/E(100)$ track unity through $T = 1000$, with $R_{1000} \in [1.000, 1.004]$ on the checked models. No energy-like systematic drift is detected.
- (iv) **Geometric sensitivity** ($A = A(\text{IC})$). Near L_4/L_5 , minor IC shifts change the residual by only $\sim 0.2\%$. A deeper excursion onto a far-equilibrium orbit raises the baseline by about 38% (uniformly for low and high N at $h = 0.05$). Thus A depends on local trajectory geometry rather than on model dimension; the h^2 law itself is preserved.

Table 7 summarizes the experimental verdict. Accepting a controlled structural $O(h^2)$ residual is preferred to forcing an artificial $O(\epsilon_{\text{mach}})$ floor by hard velocity projection, which degrades orbital fidelity without improving dimensional scalability. CR8BP–CR10BP synthetic shells are stress tests of the architecture, not Solar-System ephemerides.

Table 7: Falsification matrix for the tracking hypothesis $E_{\text{flux}} \approx A(\text{IC}) h^2$ on $N \in \{4, \dots, 10\}$.

Claimed property	Status	Metric / floor
Quadratic convergence ($E \propto h^2$)	Confirmed	Fit error $< 1\%$ on all short grids
Dimensional invariance ($E \propto N^0$)	Confirmed	Cross-model CV $\approx 0.63\%$ ($N = 4..10$)
Secular boundedness ($E \propto T^0$)	Confirmed	$R_{1000} \approx 1.00$ ($T \leq 1000$)
Universal prefactor ($A \neq A(\text{IC})$)	Falsified	A depends on orbit geometry (+38% far-case)
Asymptotic limit ($N, T \rightarrow \infty$)	Unproven	Empirical bound on tested regimes only

9 Related Work

Classical treatments of the circular restricted three-body problem and its invariants are given by Szebehely [1]. Mission design and quasi-periodic motion near libration points are surveyed by Gómez et al. [2, 10]. The present construction addresses the complementary numerical question: how to track a *moving* scalar target once an explicit perturber destroys the autonomous Jacobi integral.

Structure-preserving integration is developed systematically by Hairer, Lubich and Wanner [3] and by Blanes and Casas [4]; higher-order symplectic composition goes back to Yoshida [5]. Projection and discrete-gradient methods for first integrals are discussed by Norton et al. [6] and in the DAE literature [7, 9]. Soft residual control is conceptually related to Baumgarte stabilization [8], which introduces damping toward a constraint manifold rather than enforcing it exactly at every step—the same distinction drawn here between hard isopotential scaling and a tolerance or soft projection.

A preliminary framework note [11] introduced the two-loop Synodic Birkhoff–PEC architecture. The present paper supplies the independent numerical audit: refinement, ablation, and long-horizon tests that separate invariant residuals from trajectory fidelity and motivate disabling hard velocity scaling in the recommended CR4BP baseline.

10 Discussion

Figure 4 summarizes the central numerical trade-off: scalar residual accuracy and trajectory fidelity occupy independent axes and need not be minimized simultaneously.

Three numerical properties must be kept separate.

First, the analytical flux provides the correct moving pseudo-Jacobi target. Second, the numerical quadrature determines how accurately that target is propagated. Third, the optional scalar projection determines how closely the computed state is placed on that target.

The numerical results show that these operations need not have the same effect on trajectory fidelity.

The hard projection demonstrates the clearest example. In both autonomous and non-autonomous configurations it can drive the scalar residual to the floating-point floor. However, the CR4BP trajectory experiment shows a large positional discrepancy and a lack of trajectory convergence. Therefore,

$$E_{\text{projection}} \rightarrow O(\epsilon_{\text{mach}}) \quad (41)$$

does not imply

$$E_{\text{trajectory}} \rightarrow 0. \quad (42)$$

Why hard velocity scaling degrades the trajectory. Hard projection multiplies the entire post-step velocity by a scalar α chosen so that the instantaneous isoenergetic constraint is satisfied exactly. In the synodic frame this changes the speed without a corresponding adjustment of the direction that would be consistent with the continuous flow. The operation is therefore equivalent to an instantaneous non-physical impulse: it alters the angular momentum relative to the rotating frame and shifts the particle’s phase on the libration torus. Over many steps these impulses accumulate as a secular positional drift, even though the scalar residual remains at machine level. Soft projection ($\beta \ll 1$) or a pure tolerance guard avoids this impulse-like correction and is preferred when orbital fidelity is required.

Computational overhead. A reference implementation of the flux-only layer (Loop A trapezoidal update with no phase and $\beta = 0$) adds approximately 2% wall-clock cost relative to a pure RK4 step on the same right-hand side, confirming that the scheme is non-iterative and inexpensive. The dominant cost remains the gravitational force evaluation.

Extended horizon check. An independent flux-only run to $T = 1000$ at $h = 0.05$ and $h = 0.025$ yields $\max |\Delta J|$ of 1.51×10^{-7} and 3.76×10^{-8} respectively, statistically indistinguishable from the $T = 100$ and $T = 500$ values. No secular growth is detected on this longer interval either.

The preferred CR4BP configuration is consequently the flux-plus-phase variant with hard scaling disabled (equivalently, flux-only when the phase layer is inactive). It does not minimize the instantaneous scalar residual as aggressively as hard projection, but it retains the fourth-order trajectory convergence of the underlying RK4 propagator while providing bounded pseudo-Jacobi tracking and suppressing the trajectory drift induced by aggressive velocity closure. When a residual guard is required, the soft or threshold projection (19)–(18) is used in place of hard scaling.

The results also demonstrate why the distinction between an invariant and a moving target is not merely terminological. In the non-autonomous problem, enforcing

$$C_J(t) = C_J(0)$$

would impose the wrong physical condition. The appropriate numerical objective is instead

$$C_J(t) \approx C_{\text{target}}(t),$$

where C_{target} is generated by the continuous flux law.

11 Reproducibility and Numerical Audit

The numerical campaign is structured so that each principal claim can be associated with a separate measurable quantity.

The implementation is nondimensionalized according to

$$G(M_\odot + M_J) = 1, \quad a_J = 1, \quad n_J = 1.$$

The reported refinement grid is

$$h \in \{0.08, 0.04, 0.02, 0.01\},$$

while the long-horizon campaign uses

$$h \in \{0.05, 0.025\}.$$

All residuals are evaluated directly from the numerical state and the corresponding target quantity. No synthetic h^4 or h^2 residual model is introduced into the reported values.

Table 8: Numerical audit map.

Test	Measured quantity	Purpose
Level 1	$ F_{C,\text{analytic}} - F_{C,\text{FD}} $	Analytical flux correctness
Level 2A	$E_C(h)$	Pure flux quadrature accuracy
Level 2B	E_{proj}	Algebraic projection accuracy
CR3BP control	$\max \Delta C_J $	Autonomous baseline
CR4BP refinement	$p_J, ; p_{\text{trajectory}}$	Convergence behavior
Ablation	$\Delta r, ; \Delta J$	Component localization
Long horizon	$\Delta J(T)$	Secular-growth audit

12 Limitations

The conclusions are restricted to the validated CR3BP control and Sun–Jupiter–Saturn CR4BP configuration and to the integration horizons explicitly tested.

No universal trajectory-preservation claim is made for the hard scalar projection. Observed residual rates—approximately first-order on the short refinement grid and near second-order on the long-horizon h -pair—are treated as empirical properties of the complete scheme in those regimes and are not identified with the formal order of an isolated quadrature rule.

A flux-only scalability ladder through CR10BP (Section 8), including an $N \times h \times T$ matrix and multi-IC checks, supports the empirical law $E_{\text{flux}} \approx A(\text{IC}) h^2$ with A nearly independent of N and T on the tested domain ($N \leq 10, T \leq 1000$). This does not constitute a proof for arbitrary CRNBP or infinite time; further bodies and horizons still require independent campaigns.

Similarly, the absence of detectable secular growth up to $T = 500$ should not be interpreted as a mathematical proof of boundedness for all times.

The distinction between scalar constraint accuracy and orbital fidelity should therefore remain explicit in any subsequent extension of the method.

13 Conclusion

A non-iterative flux-tracked geometric projection architecture has been formulated and numerically audited in autonomous CR3BP and non-autonomous CR4BP settings.

The central construction replaces the inappropriate assumption of a constant Jacobi integral in the non-autonomous problem with an explicitly integrated pseudo-Jacobi flux and a corresponding moving target manifold.

The short-horizon CR4BP refinement campaign establishes approximately first-order scaling of the flux-only residual on $h \in \{0.08, 0.04, 0.02, 0.01\}$:

$$E_J(h) = O(h) \quad (\text{empirical on that grid}),$$

while the underlying trajectory retains approximately fourth-order convergence:

$$E_{\text{trajectory}}(h) = O(h^4).$$

The measured residual decreases from 2.75×10^{-6} at $h = 0.08$ to 3.49×10^{-7} at $h = 0.01$ (consecutive orders near unity). Independently, the long-horizon h -refinement at fixed T yields an approximately fourfold residual drop when h is halved ($p_{\text{long}} \approx 2$ on $h \in \{0.05, 0.025\}$). These rates are empirical properties of the complete scheme in the tested regimes.

The long-horizon campaign provides an additional independent result. At $T = 100$ and $T = 500$, the residuals for identical step sizes are effectively unchanged:

$$1.57 \times 10^{-7} \rightarrow 1.58 \times 10^{-7}$$

for $h = 0.05$, and

$$3.92 \times 10^{-8} \rightarrow 3.93 \times 10^{-8}$$

for $h = 0.025$.

No detectable secular growth is therefore observed over the tested interval up to $T = 500$.

The ablation experiment demonstrates that hard velocity scaling, although capable of enforcing the scalar constraint at approximately the double-precision floor, can substantially degrade orbital fidelity. The full PEC configuration produces a positional deviation approximately 11.4 times larger than the flux-plus-phase configuration with scaling disabled, while the hard projection has a positional discrepancy of approximately 4.3 relative to the finest pure-RK4 reference.

The validated conclusion is therefore not that the smallest scalar constraint residual defines the best numerical solution. Rather, the preferred CR4BP configuration is the one that tracks the moving pseudo-Jacobi target—with optional symmetric phase correction and without hard velocity scaling—while preserving the convergence properties of the underlying trajectory integrator. Drift is controlled by separating flux tracking from aggressive isopotential projection, not by forcing the scalar residual to machine precision at every step.

The principal methodological result is consequently the separation

$$\boxed{\text{moving-manifold tracking} \neq \text{instantaneous scalar projection} \neq \text{trajectory fidelity}.}$$

This separation provides a basis for further extensions to higher-dimensional restricted many-body configurations, provided that each extension is subjected to the same independent analytical, quadrature, projection, refinement, ablation, and long-horizon validation protocol.

Data and Code Availability

A self-contained Python reference implementation of the CR3BP/CR4BP flux-tracked projection layer, including pure RK4, flux-only and hard projection modes, refinement drivers, ablation diagnostics, and long-horizon tests, is maintained by the author.

The numerical tables reported here are reproducible from the stated nondimensionalization and numerical protocol.

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